GLOBAL NAVIGATION SATELLITE SYSTEMS

Relative static GNSS measurements and data processing

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Theoretical Background:

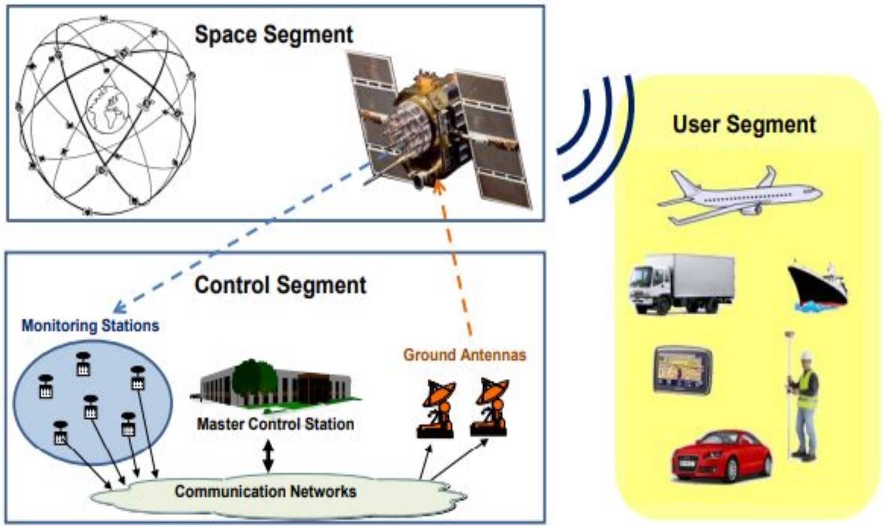
GNSS=Global Navigation Satellite System

Goals :

- Real time navigation
- Anywhere in the world
- Under any weather conditions
- Precise timing(Clock Synchronisation)

Three Segments:

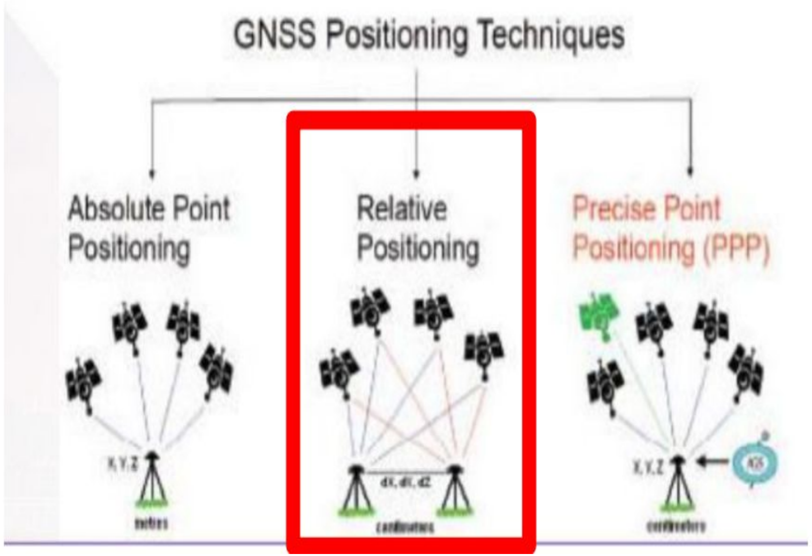
- Space segment
- Control segment
- User Segment



Source :GNSS DATA PROCESSING BOOK

Relative Positioning :

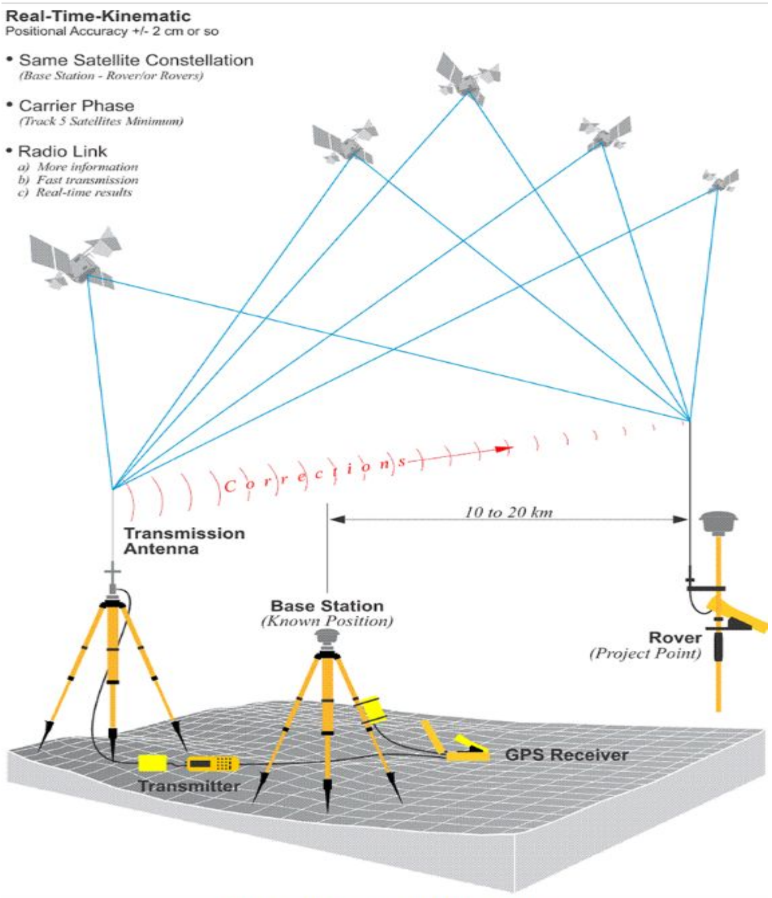
- **Own Master Station** :one point acts as a reference station(master) and its position is known while the position of the other point(s) is calculated relative to the reference station.
- **Virtual Reference Station (VRS):**In this method,the data are generated from multiple reference stations.The system are called SAPOS (Satellite-based Augmentation Service for Precise Orbit and Positioning) is used to generate VRS.



Source :GNSS DATA PROCESSING BOOK

Relative Positioning :

- In both methods, the observations used in relative positioning are the time of flight (TOF) of the signals from the satellites.
- It's more precise than absolute positioning because it removes errors that are common to both reference and rover receivers, such as satellite clock errors and atmospheric errors.



Relative (Differential) with RTK Positioning

Source: GPS for Land Surveyors

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Static GNSS Measurement:

- Taking the observations over a period of time(minutes to hours).
- The purpose of this measurement to improve the accuracy of the position solution by averaging out errors.

Errors occurred due to several reasons like:

- satellite clock errors
- atmospheric errors
- receiver noise

Measurement Process:

Points to check about:

Short Baseline:

- Post-processing
- static application
- relative GNSS measurements
- carrier-phase observations
- **Broadcast orbits from navigation message**
- **>1-2 hours for observation**
- **One- or two-frequency measurements possible**
- Equipment used :tripod,antenna,receiver(normal)

Long Baseline:

- Post-processing
- static application
- relative GNSS measurements
- carrier-phase observations
- **IGS final orbits**
- **>8-12 hours for observation**
- **two-frequency measurements**
- Equipment used :Heavy tripod,antenna height,receiver(high-end)

Measurement Process:

CARRIER PHASE MEASUREMENT

ambiguities are time independent!

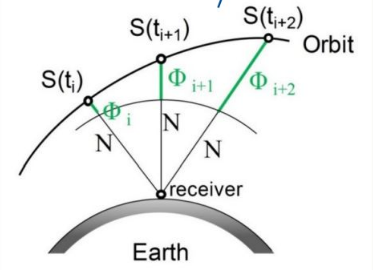
$$\phi = 1/\lambda\left[r + I_{\phi} + T_{\phi}\right] + c/\lambda\left[\underbrace{\delta t_{u} - \delta t^{s}}_{\text{Clock biases}}\right] + N + \varepsilon_{\phi}$$

Geometric range

ionospheric and tropospheric Refraction

Clock biases

noise + other effects



carrier-phase observations are more precise than pseudorange observations

G1 Point of Solution accuracy for GPS only (Ground Track):

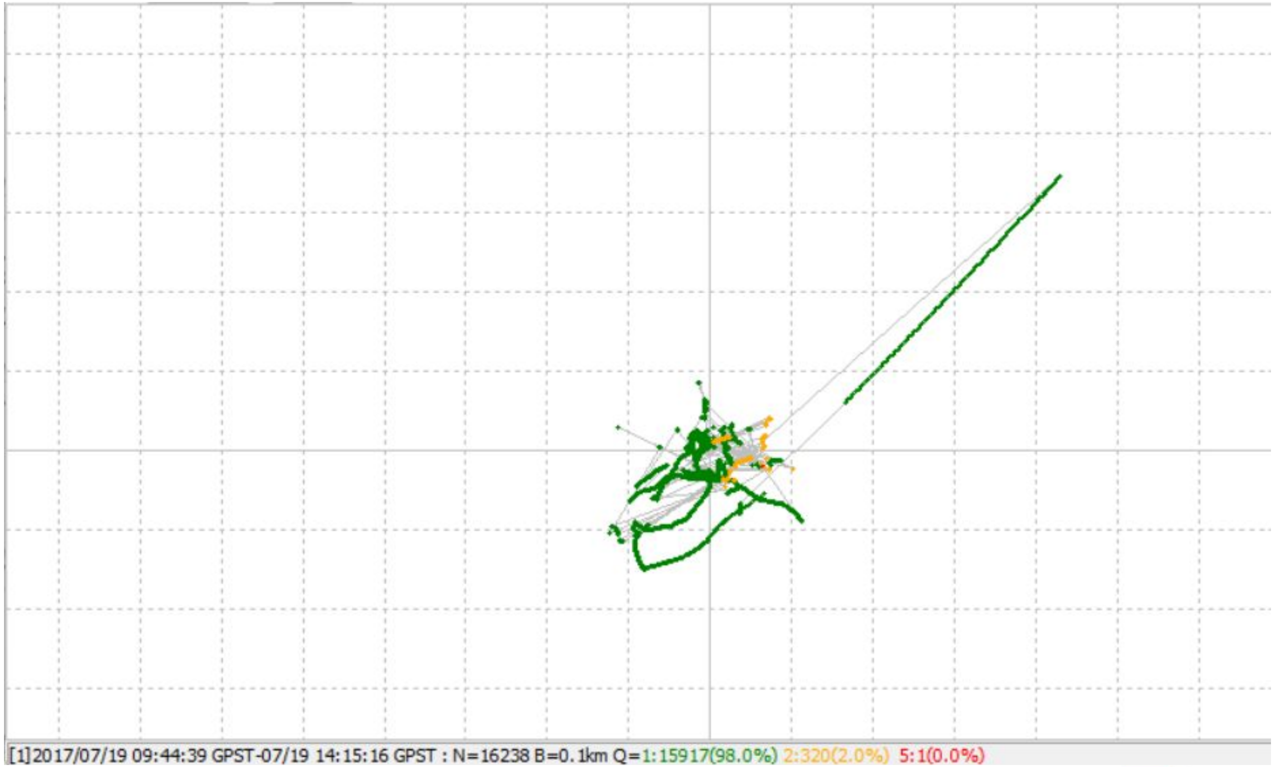
Static Method

Frequency-L1

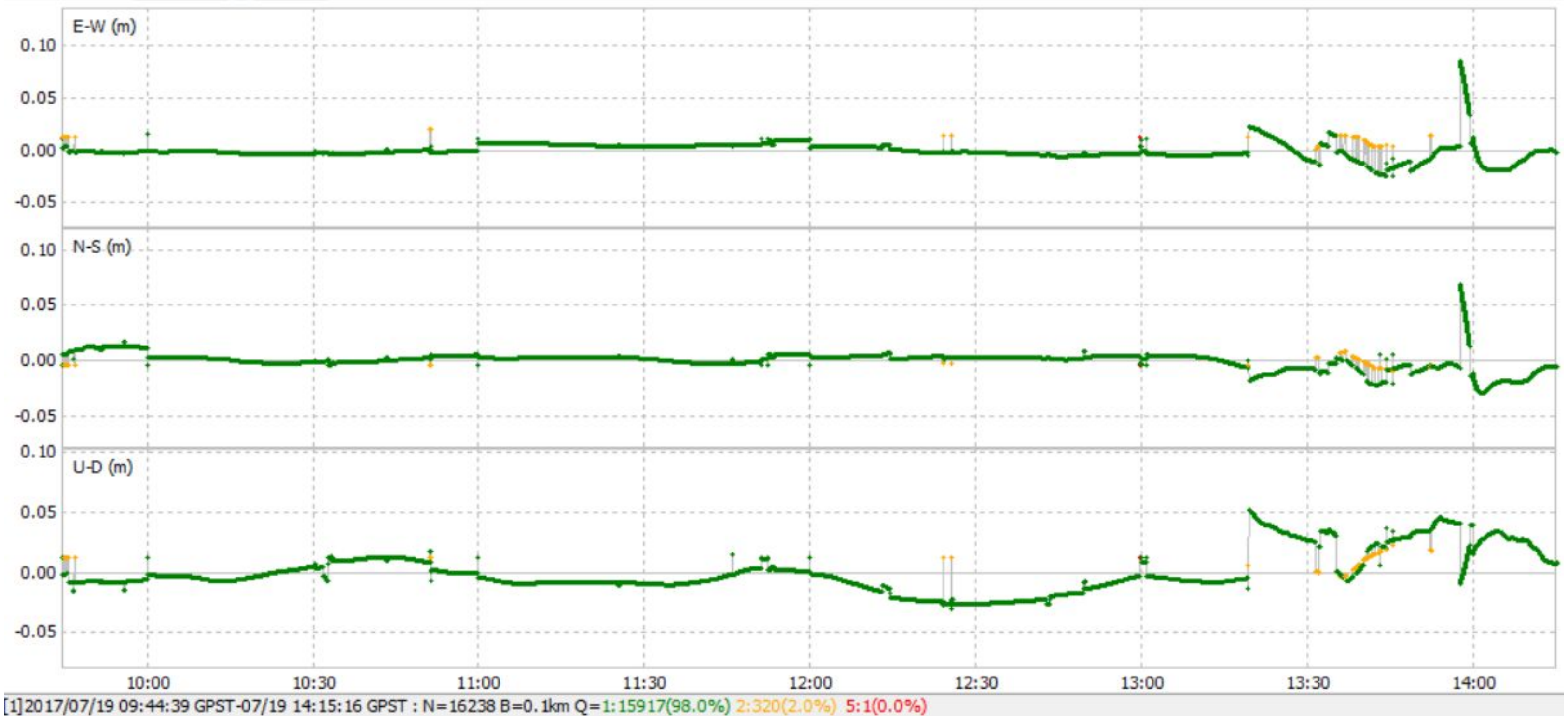
Elevation mask-15

Solutions

No. of Satellites -7



G1 Point Static Method (Position):



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G1 Point of Solution accuracy for GPS only (Ground Track)

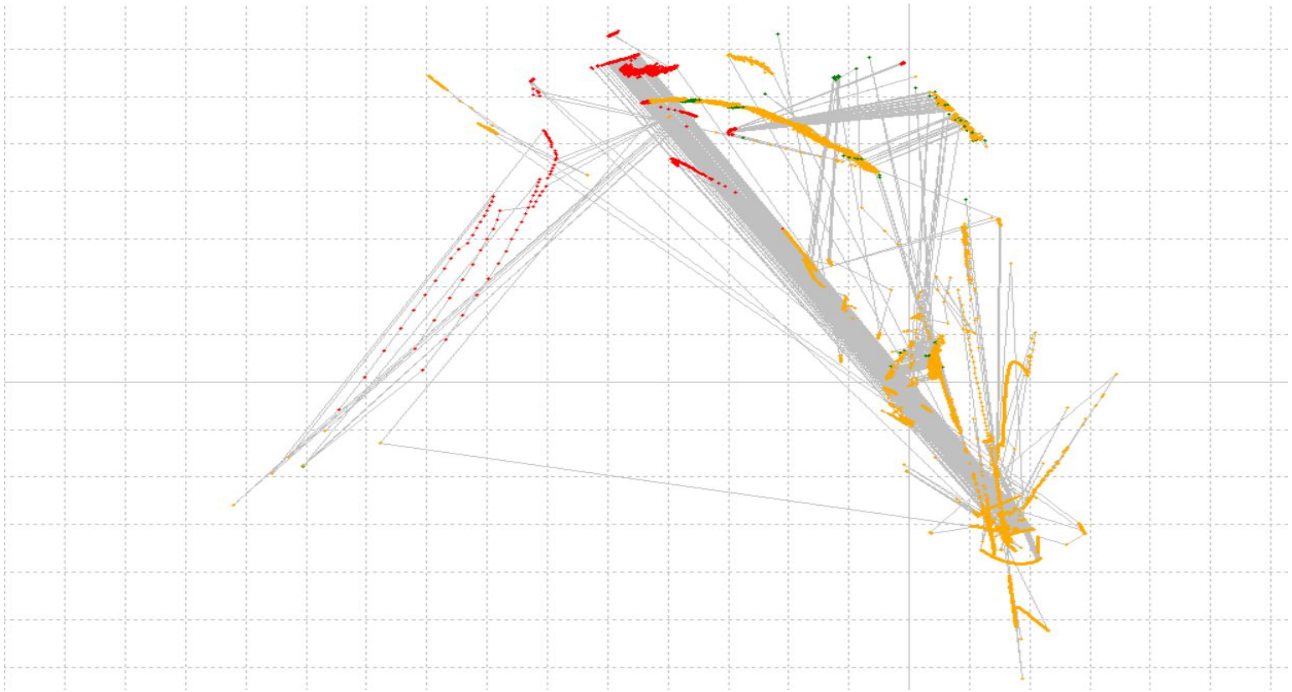
Kinematic Method

Frequency-L1

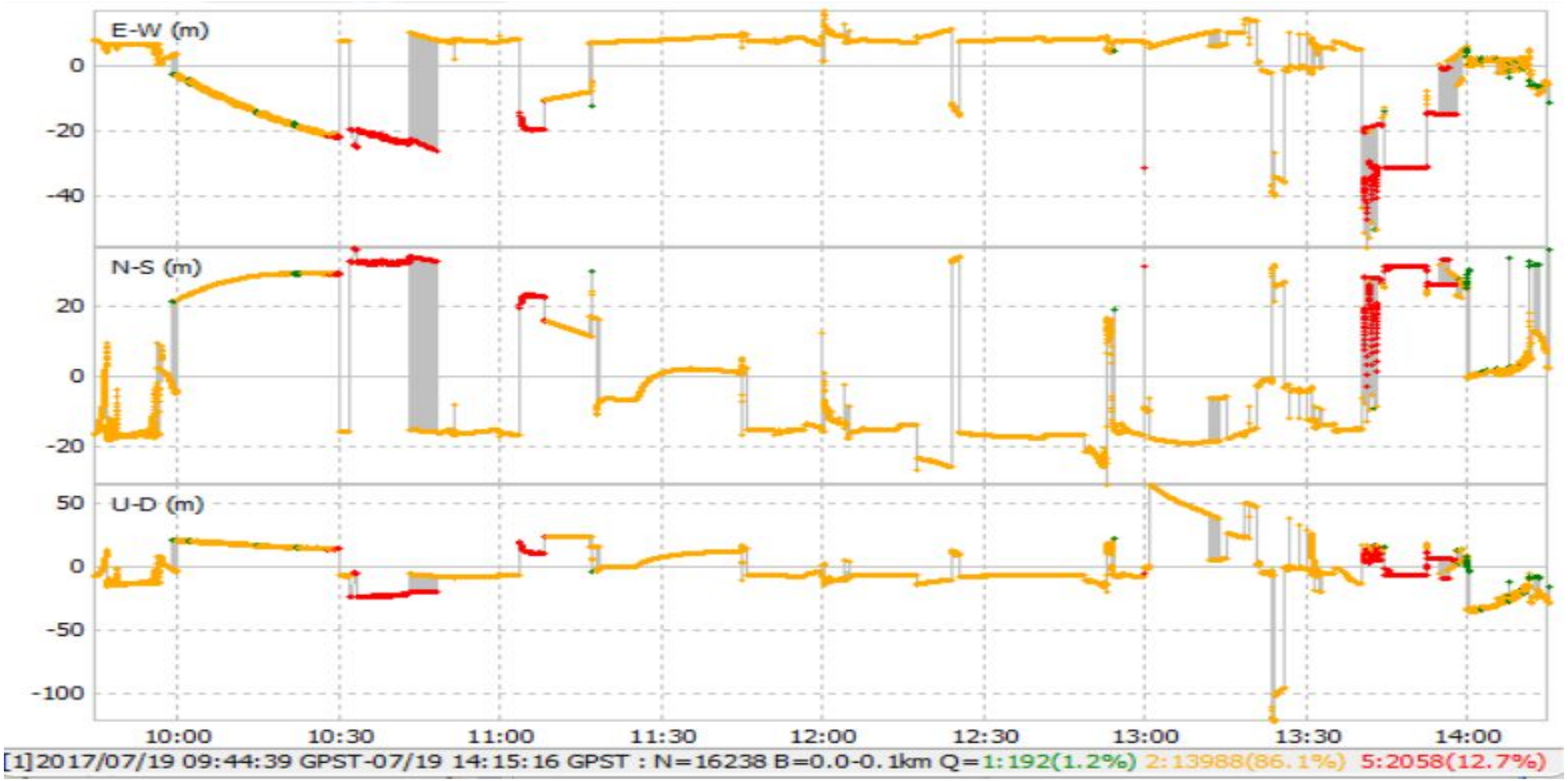
Elevation mask-15

Solutions

No. of Satellites -7



G1 Point Kinematic Method (Position):



G1 Point of Solution accuracy for GPS only (Ground Track):

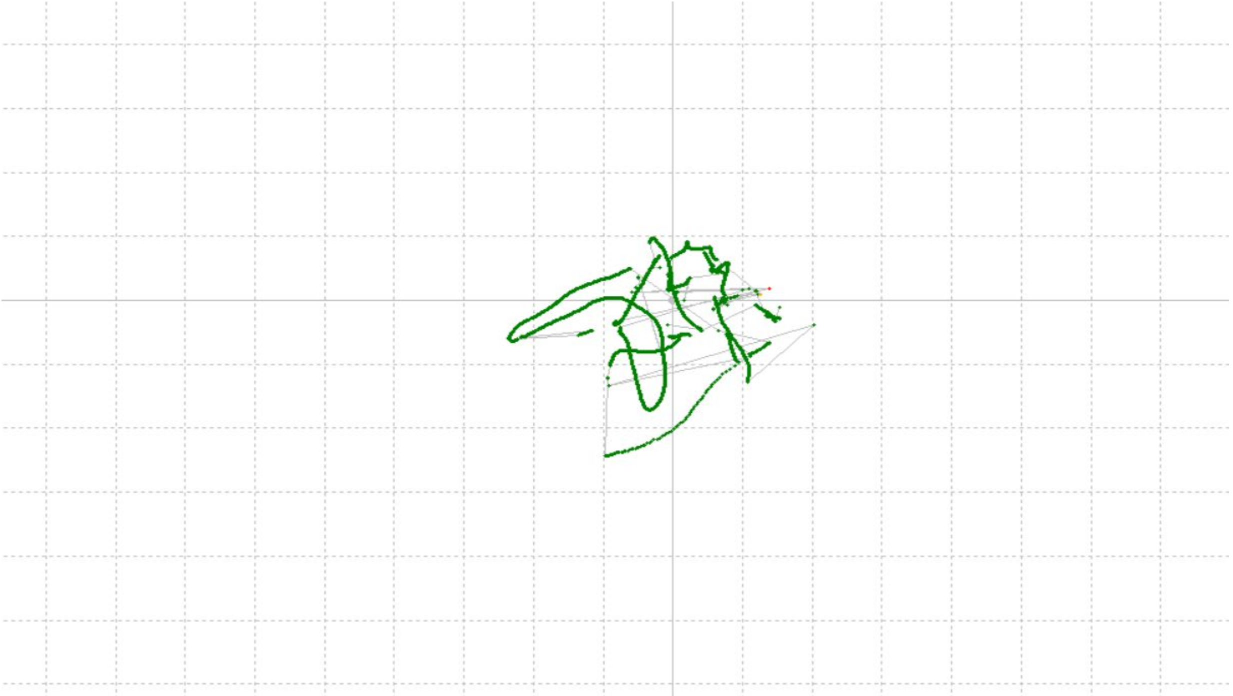
Static Method

Frequency-L1+2

Elevation mask-30

Solutions

No. of Satellites -4



G1 Point Static Method (Position):

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G1 Point of Solution accuracy for GPS only (Ground Track)

Kinematic Method

Frequency-L1+2

Elevation mask-30

Solutions

No. of Satellites -4

G1 Point Static Method (Position):

G2 Point of Solution accuracy for GPS only (Ground Track):

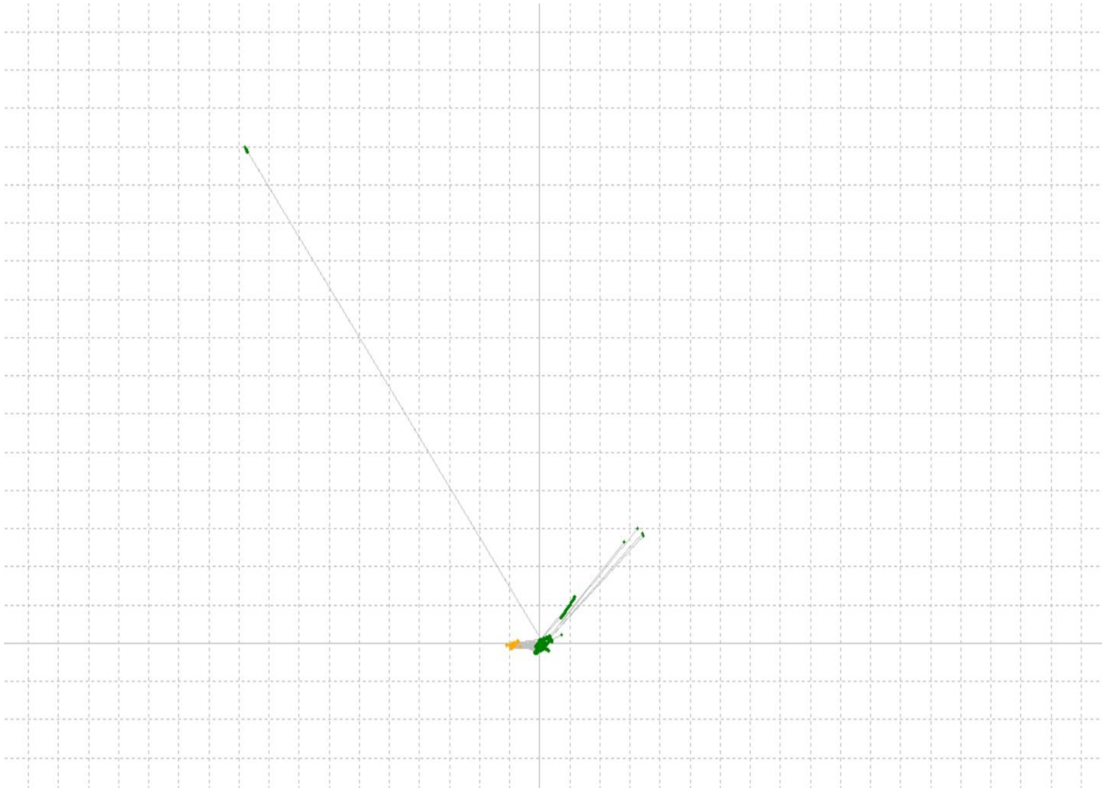
Static Method

Frequency-L1

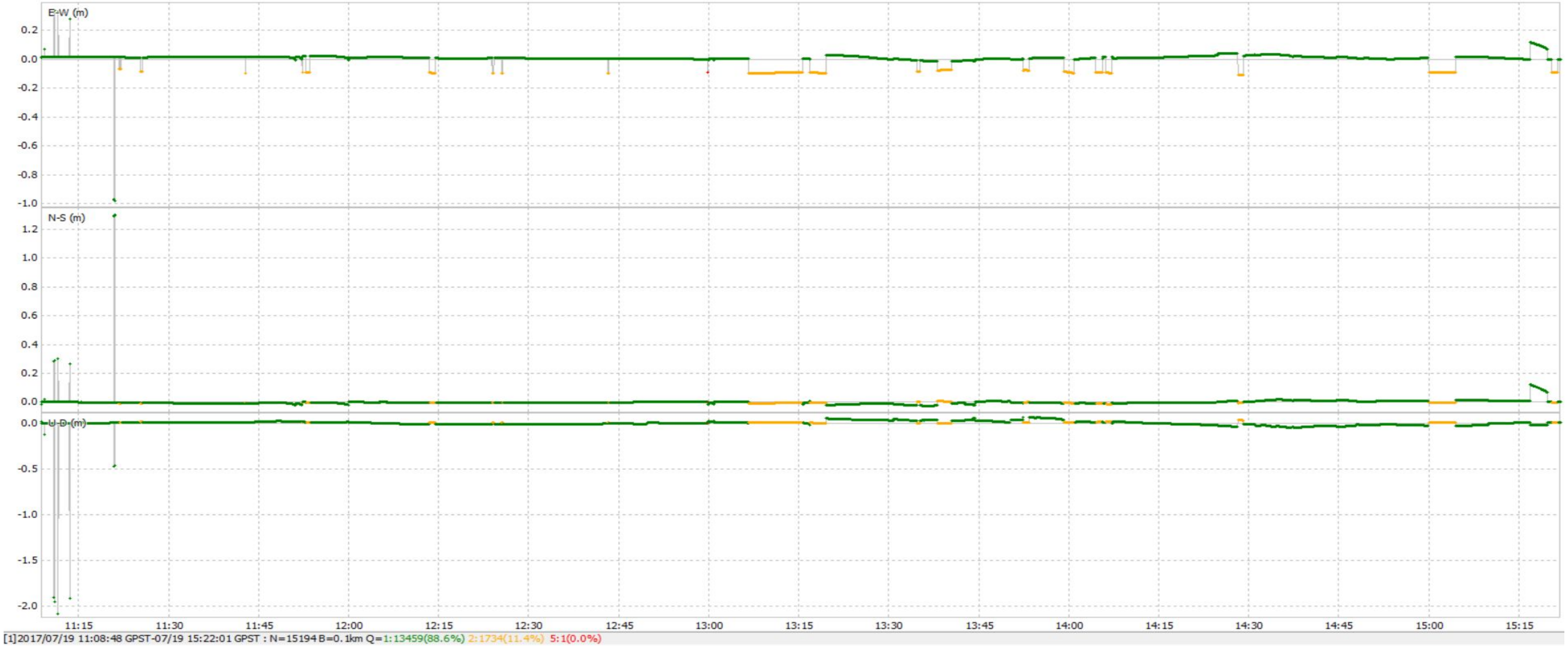
Elevation mask-15

Solutions

No. of Satellites -8



G2 Point Static Method (Position):



G2 Point of Solution accuracy for GPS only (Ground Track)

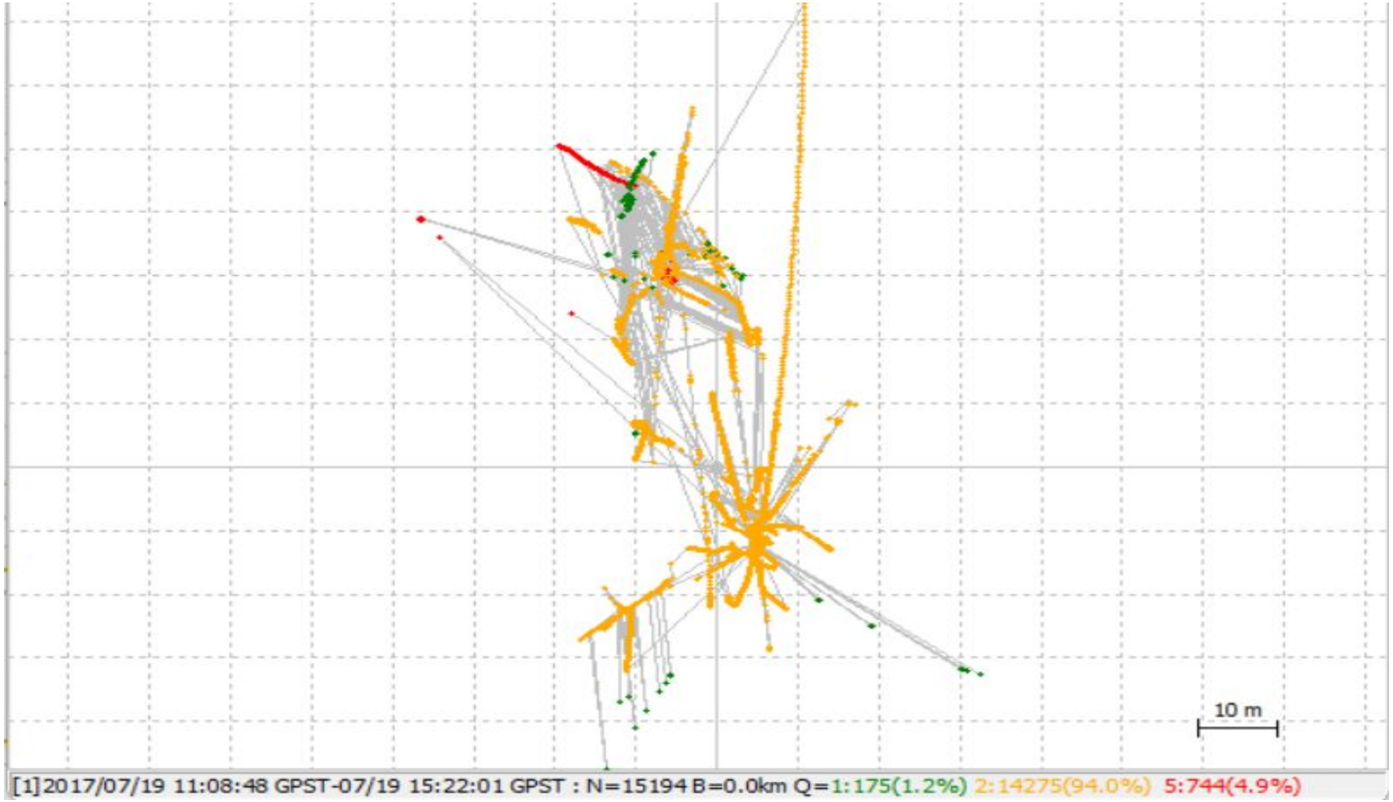
Kinematic Method

Frequency-L1

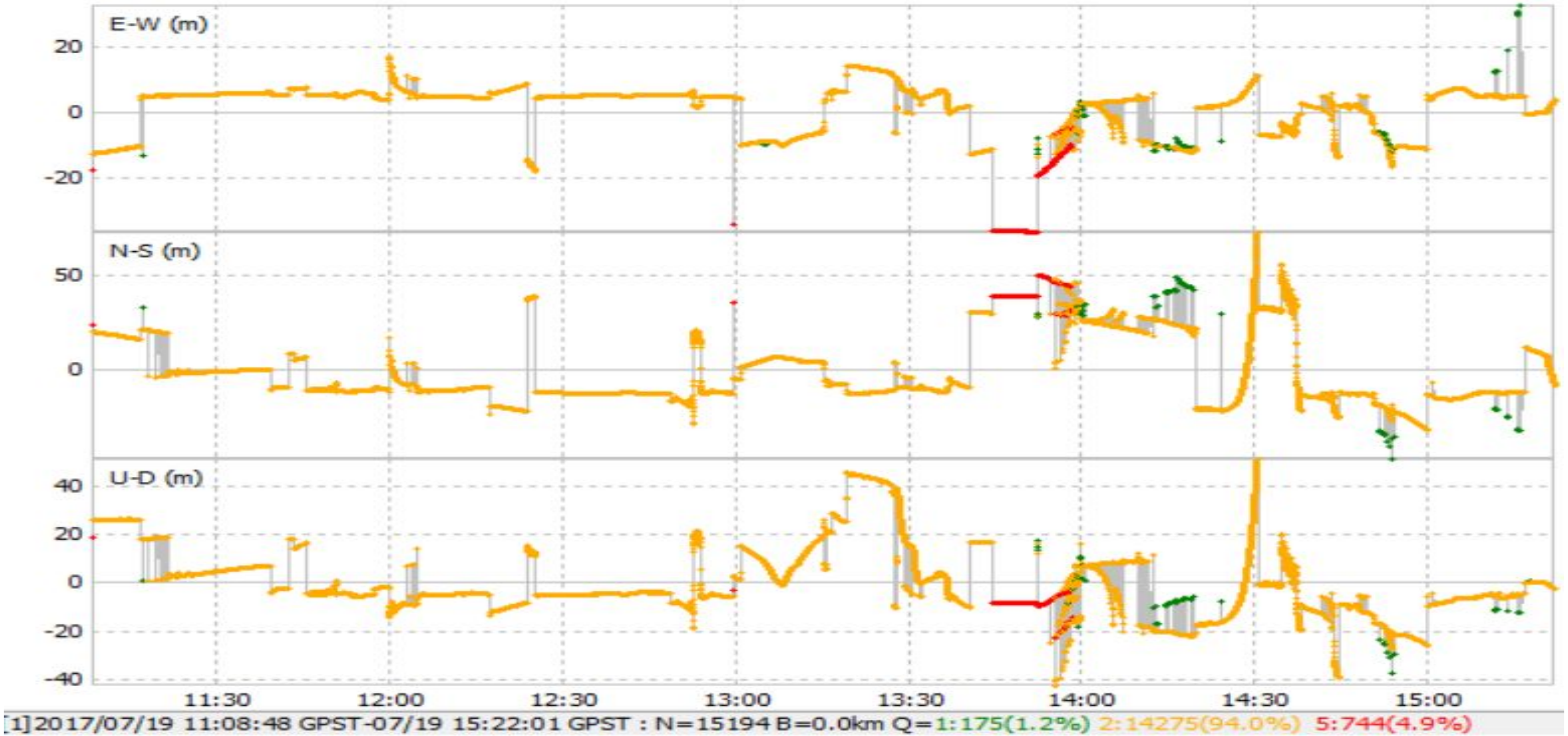
Elevation mask-15

Solutions

No. of Satellites -5



G2 Point Kinematic Method (Position):



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G4 Point of Solution accuracy for GPS only (Ground Track):

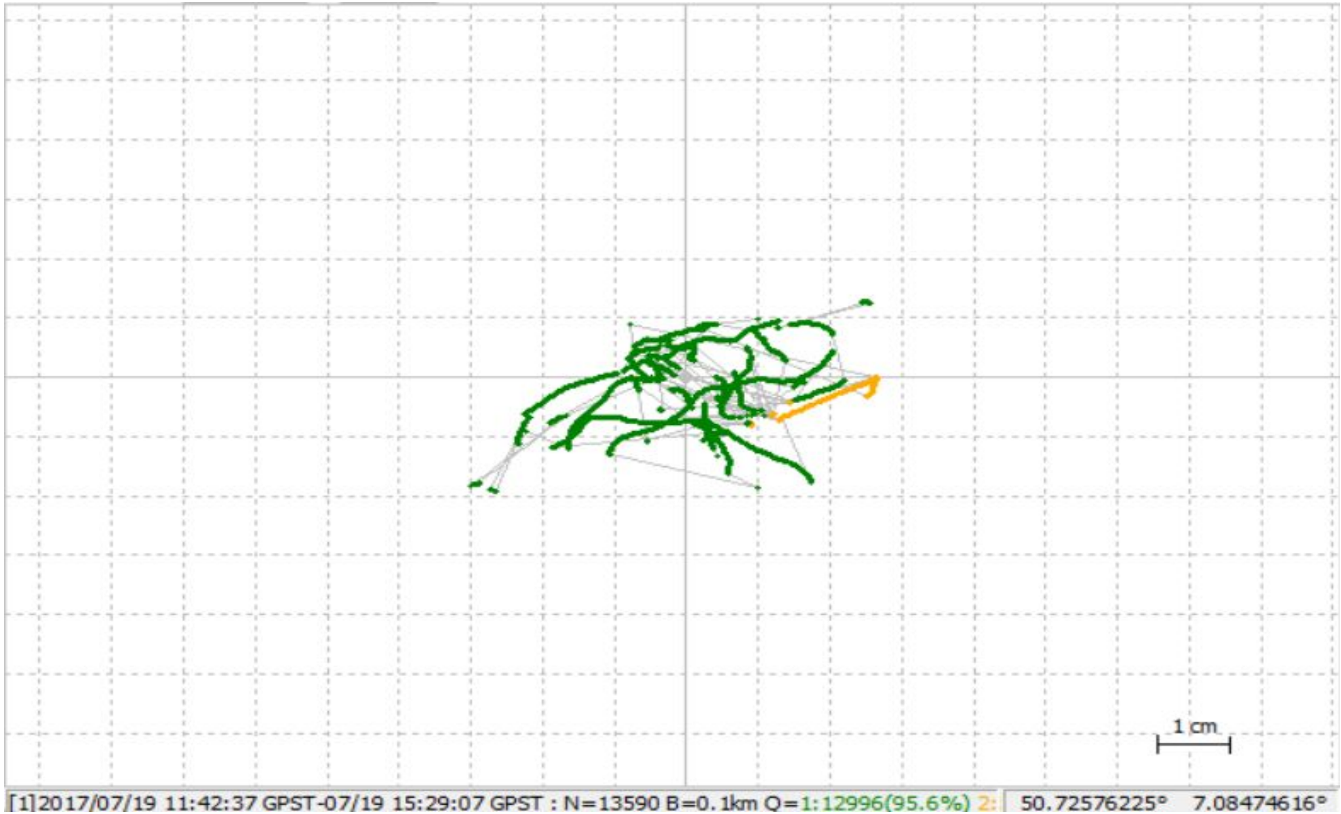
Static Method

Frequency-L1

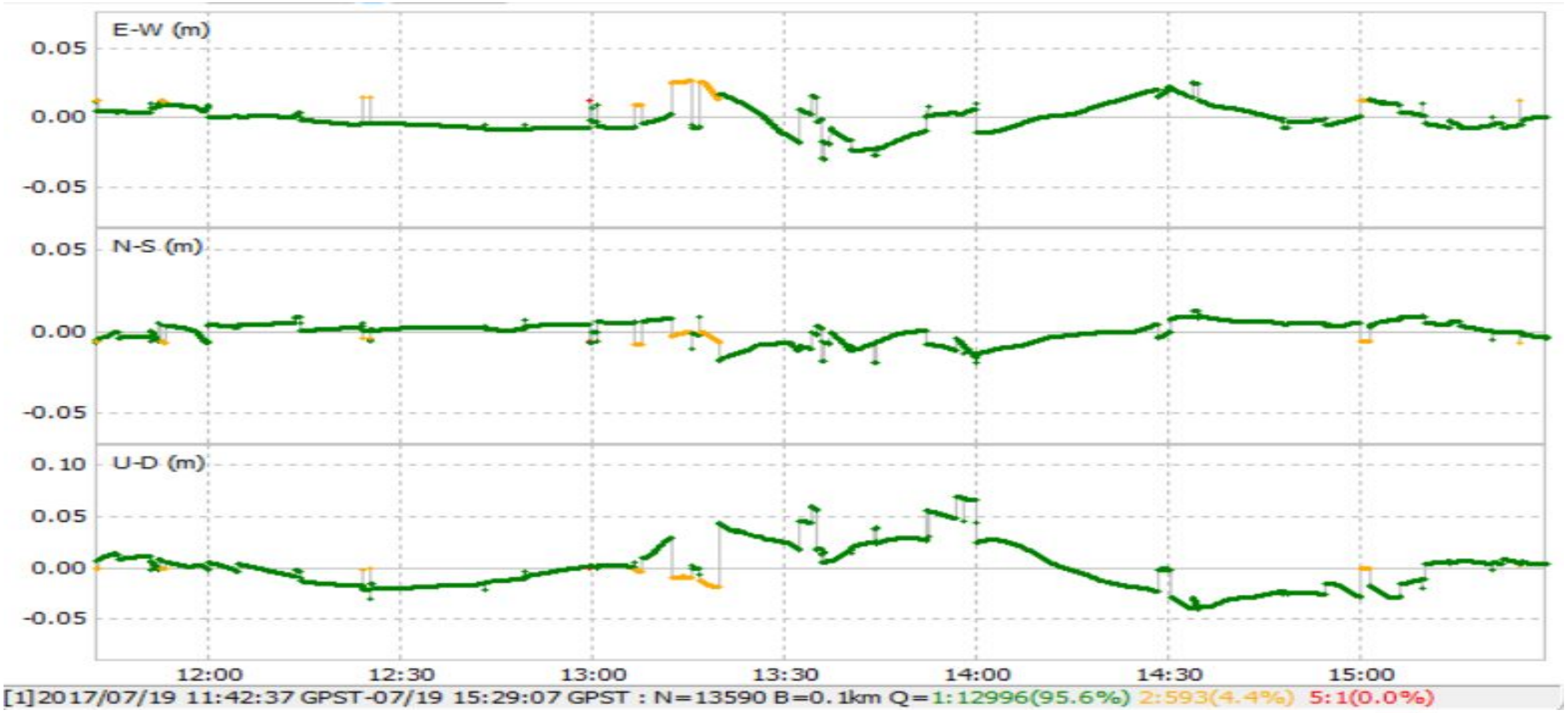
Elevation mask-15

Solutions

No. of Satellites -8



G4 Point Static Method (Position):



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G4 Point of Solution accuracy for GPS only (Ground Track)

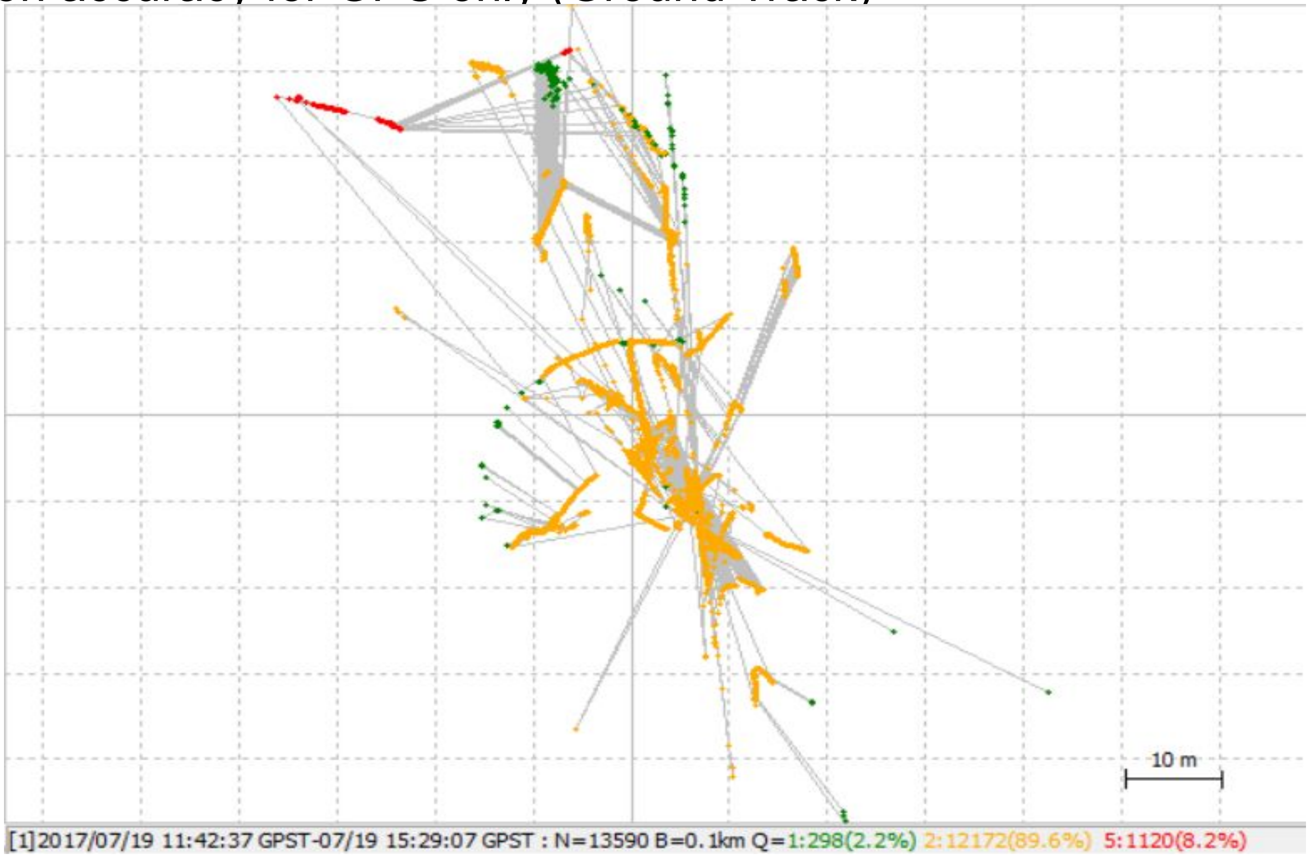
Kinematic Method

Frequency-L1

Elevation mask-15

Solutions

No. of Satellites -8



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G4 Point Kinematic Method (Position):

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G6 Point of Solution accuracy for GPS only (Ground Track):

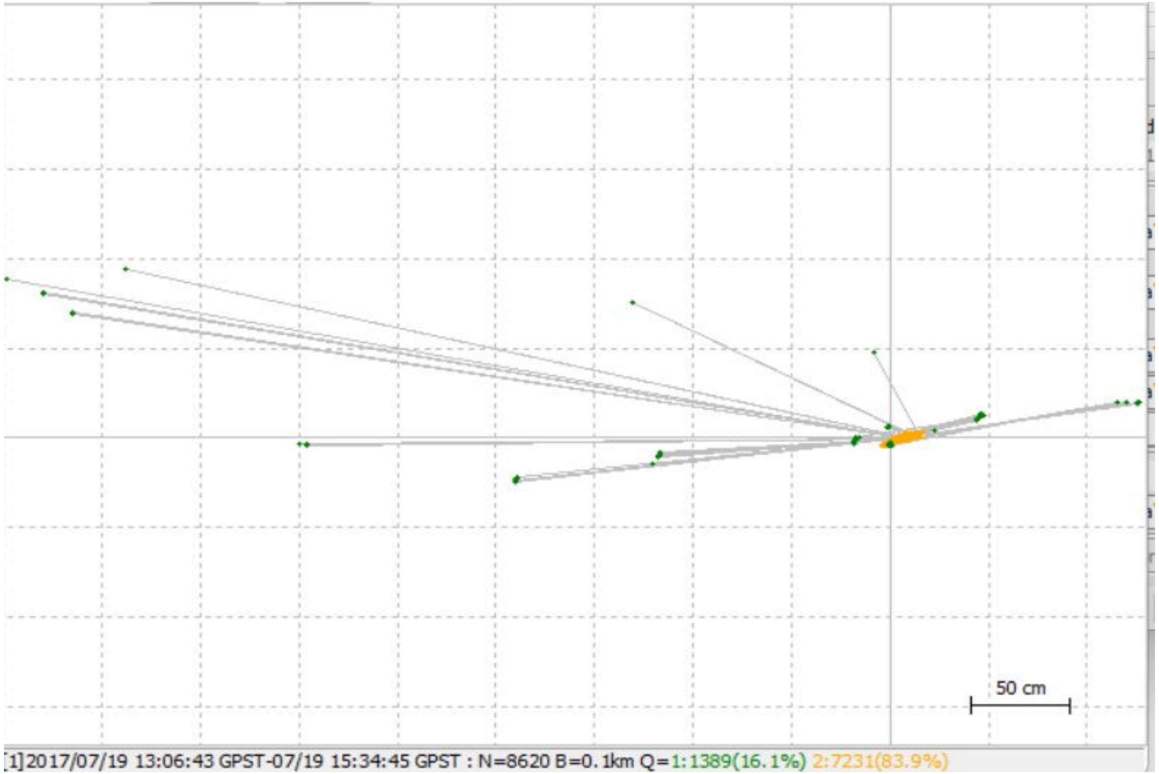
Static Method

Frequency-L1

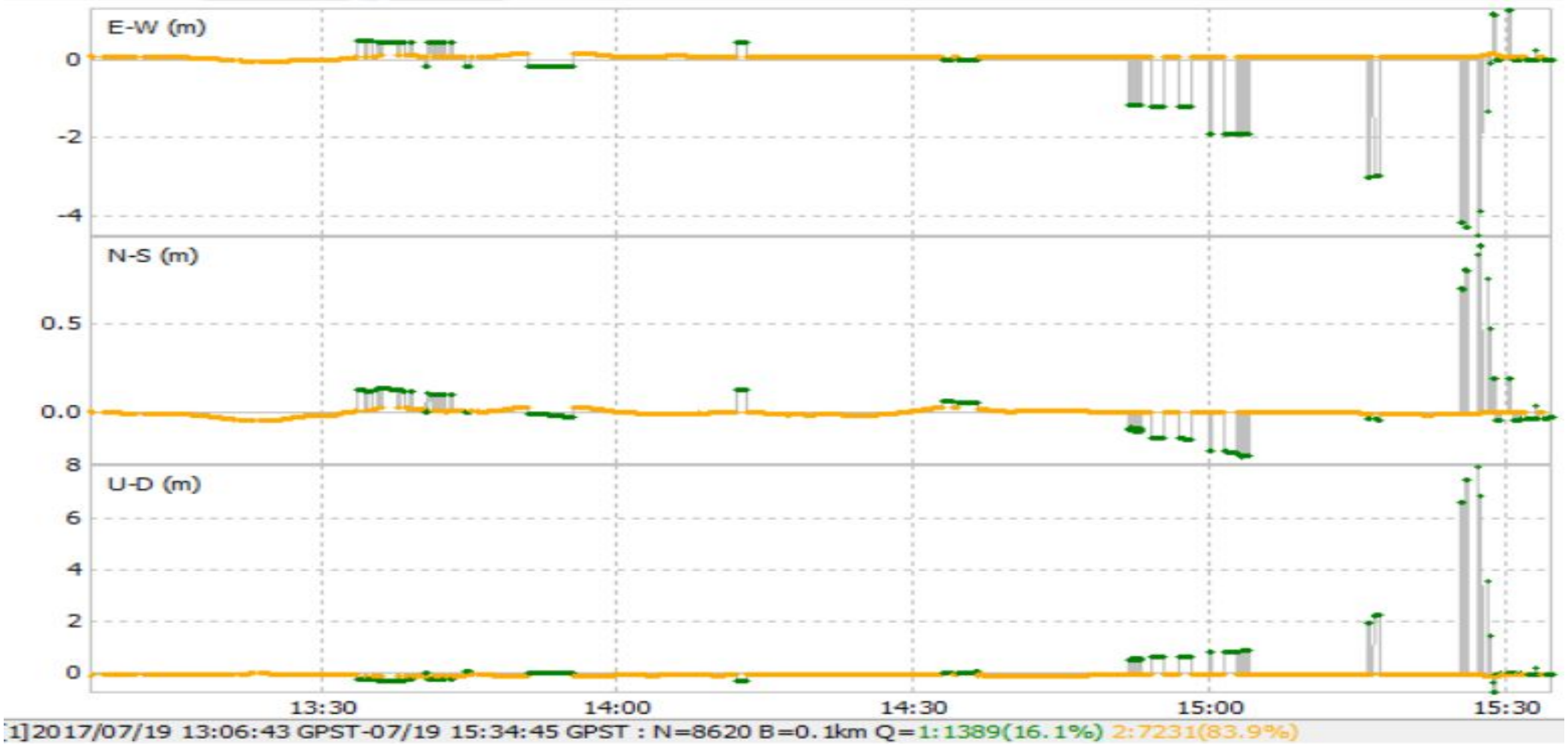
Elevation mask-15

Solutions

No. of Satellites -7



G6 Point Static Method (Position):



G6 Point of Solution accuracy for GPS only (Ground Track)

Kinematic Method

Frequency-L1

Elevation mask-15

Solutions

No. of Satellites -6

G6 Point Kinematic Method (Position):

G7 Point of Solution accuracy for GPS only (Ground Track):

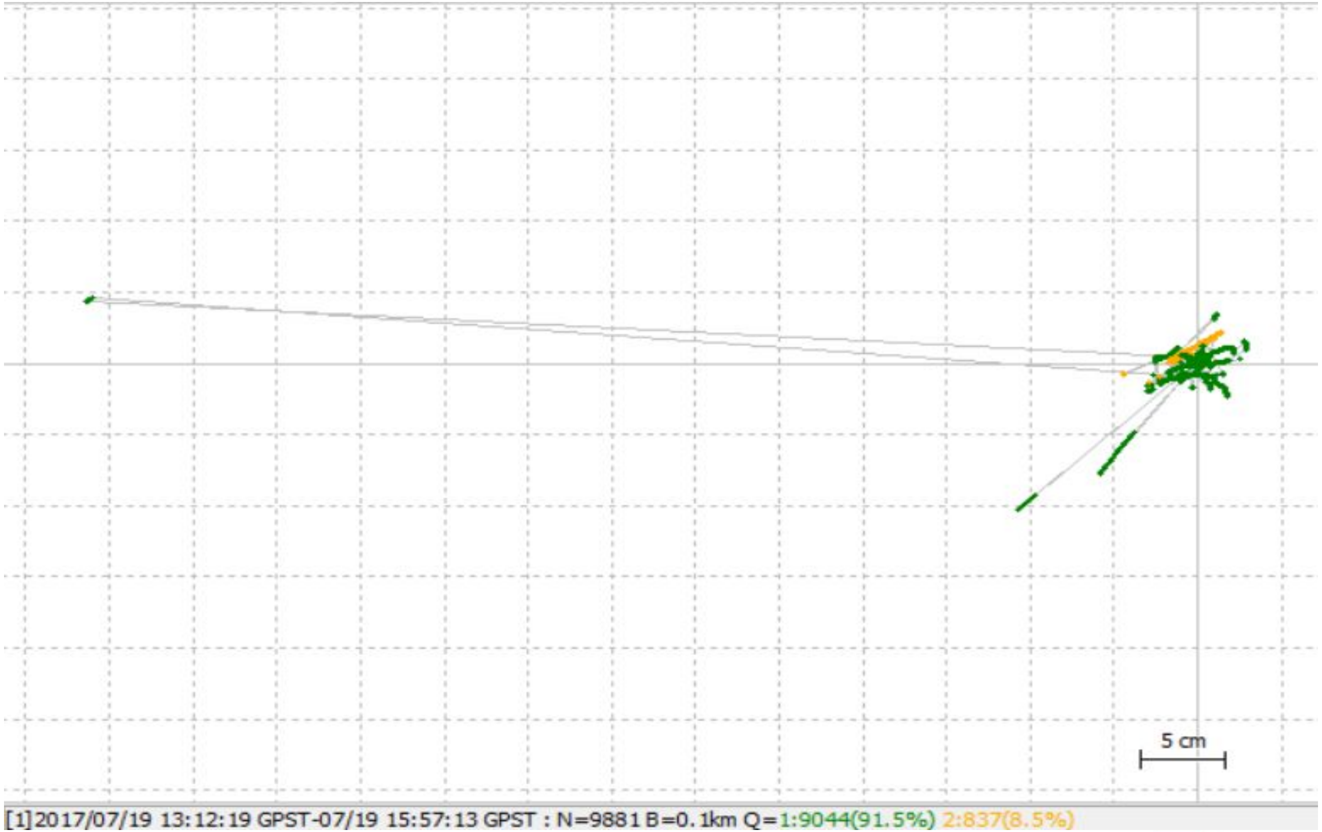
Static Method

Frequency-L1

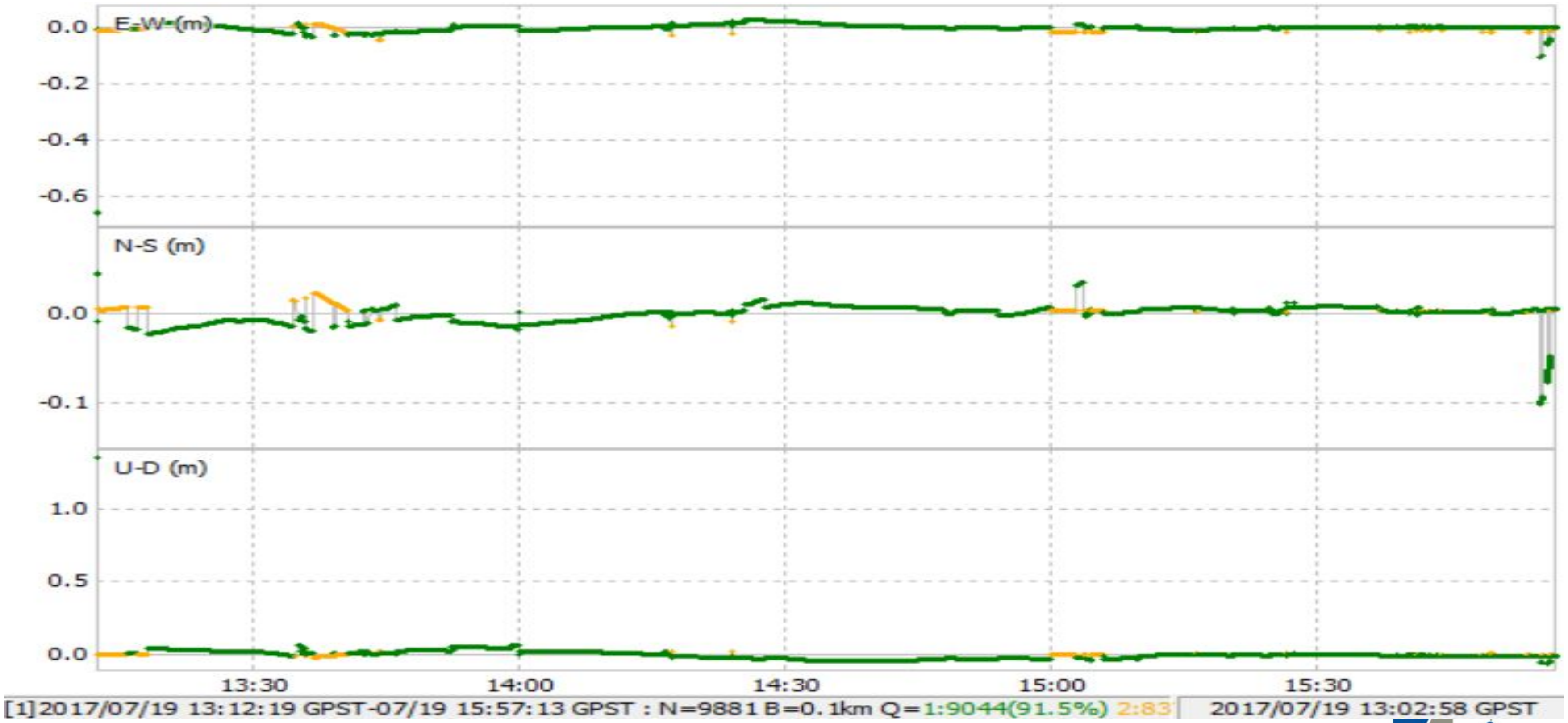
Elevation mask-15

Solutions

No. of Satellites -8



G7 Point Static Method (Position):



G7 Point of Solution accuracy for GPS only (Ground Track)

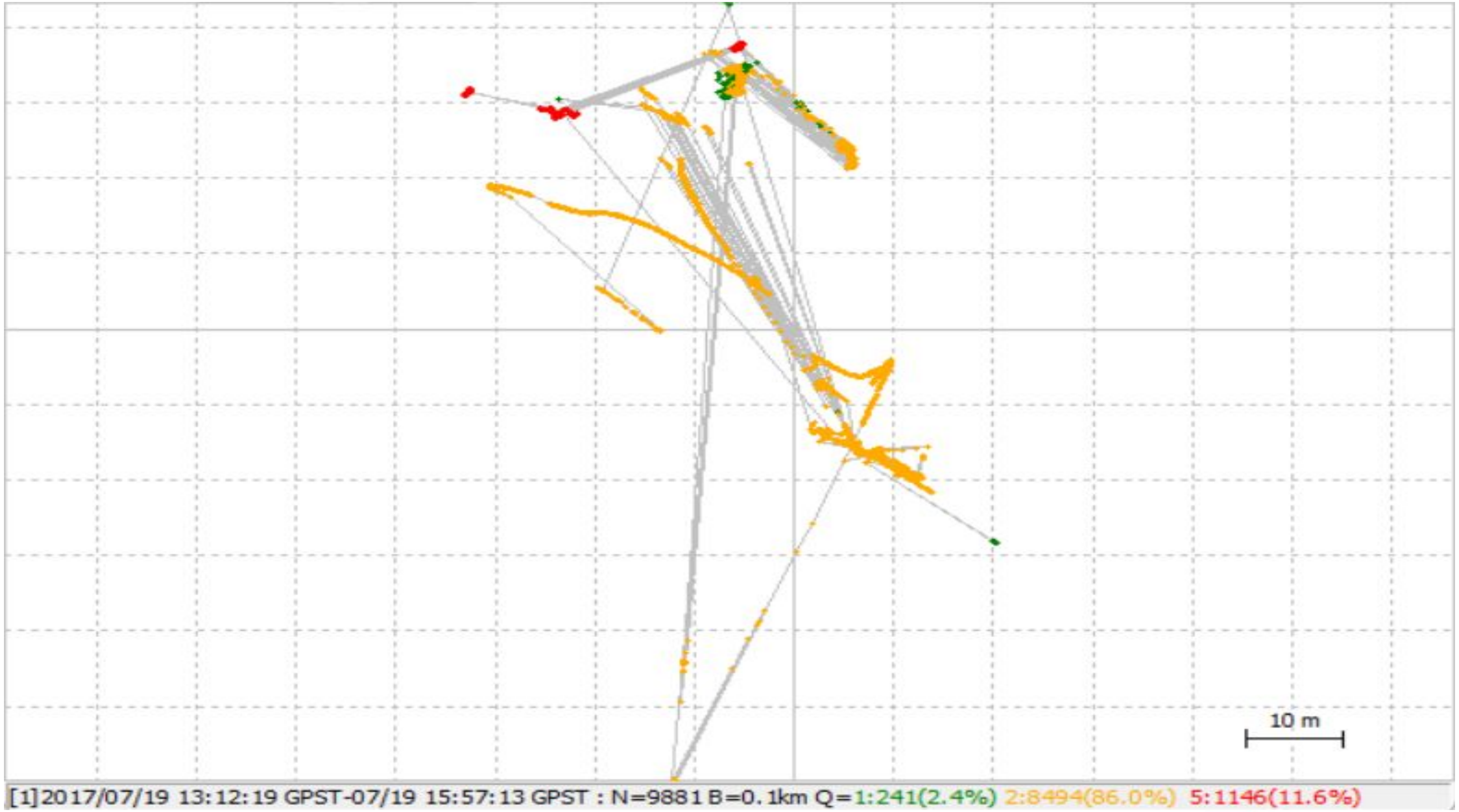
Kinematic Method

Frequency-L1

Elevation mask-15

Solutions

No. of Satellites -8



G7 Point Kinematic Method (Position):

G8 Point of Solution accuracy for GPS only (Ground Track):

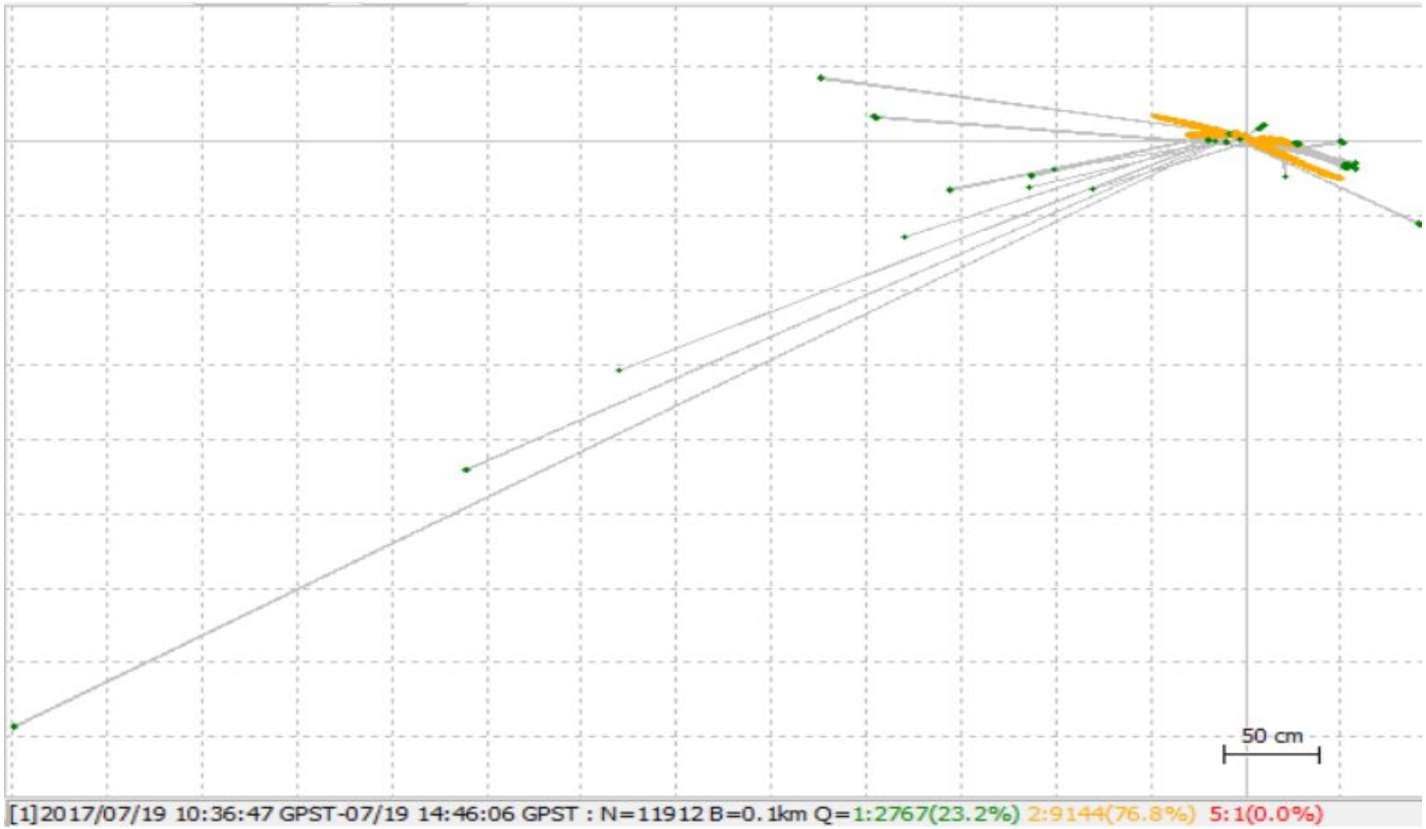
Static Method

Frequency-L1

Elevation mask-15

Solutions

No. of Satellites - 6



G8 Point Static Method (Position):

G8 Point of Solution accuracy for GPS only (Ground Track)

Kinematic Method

Frequency-L1

Elevation mask-15

Solutions

No. of Satellites -6

G8 Point Kinematic Method (Position):

Evaluation:

- In static method , the points doesn't have a single point solution.
- In static method ,the point G6,G7,G8 have more float solution because a buildings are surrounded by them.
- In Kinetic method, all points accuracy have more a float solution because a number of satellite is less.

Thanking You!!!